

ENAA CONNECT

Software Requirements Specification

Campus community platform

PLANNING BRIEF

Scope / roles / requirements / use cases / user flows

Project Overview

A concise specification for building the EnaaConnect website from the beginning.

Purpose

EnaaConnect shall provide an authenticated campus space where ENAA members share posts, maintain profiles, discover approved events, and participate in campus activities. Administrators shall review submitted events before publication.

Problem and objective

Campus activity and community updates need a single place that connects people and events. The objective is to provide a clear, role-aware website supported by persistent accounts, posts, comments, events, and notifications.

In scope

Area	Required capability
Accounts	Registration, login, current-user access, logout, and password reset by email.
Community	Profiles, posts with an optional image, likes, and comments.
Events	Submission, review, approved-event discovery, participation, and participant lists.
Activity	Notifications generated by selected post and event actions.

Scope boundary	Clubs, messaging, trips as a separate module, search, and bookmarking are not requirements of this specification.
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Reading this document

This document defines the target behavior and boundaries for the first implementation.

Users and Roles

Two account roles define protected actions. A visitor is the only unauthenticated user state.

Actor	Purpose	Permitted scope
Visitor	Access the system	Register as a student; log in; request or complete a password reset.
Student	Join the campus community	Use protected content; maintain own profile; create posts/events; join or leave events; update own posts.
Administrator	Review submitted events	Use protected content; view pending events; approve or reject pending events.

Access matrix

Action	Visitor	Student	Admin
View protected feed and events	No	Yes	Yes
Create post or submit event	No	Yes	Yes
Update own post	No	Yes	No
Join or leave event	No	Yes	No
Review pending events	No	No	Yes
Edit own profile	No	Yes	Yes

Important rule Registration creates a student account. This specification does not define an administrator registration flow.

All protected actions must be checked by the server. Hiding a control in the browser is supplementary and does not grant or remove authority.

Main Features: Accounts and Profiles

Account access

ID	Requirement
AC-01	The system shall register an account using full name, ENAA email, password, and password confirmation.
AC-02	The system shall authenticate valid credentials and provide a token for protected requests.
AC-03	The system shall return the current signed-in user's safe profile data.
AC-04	The website shall clear its stored token when the user signs out.
AC-05	The system shall support a password-reset request by email and a time-limited reset link.

Profile management

ID	Requirement
PR-01	A signed-in user shall be able to view and edit their own allowed profile fields.
PR-02	A signed-in user shall be able to upload one profile picture.
PR-03	A signed-in user shall be able to retrieve a student's public profile by ID.
PR-04	The profile view shall show posts authored by the signed-in user.

Profile fields The editable profile fields are full name, field of study, academic year, biography, and interests. The picture is uploaded separately.

Password reset and public-profile retrieval are required capabilities. This brief does not prescribe separate website screens for them.

Main Features: Posts and Comments

Post feed

ID	Requirement
PO-01	A signed-in user shall see posts in newest-first order.
PO-02	A signed-in user shall be able to publish text, one image, or both.
PO-03	A signed-in user shall be able to like or unlike a post.
PO-04	A student shall be able to change the text of their own post.
PO-05	A post owner shall be able to delete their post; its comments shall also be removed.

Comments

ID	Requirement
CO-01	A signed-in user shall be able to add a comment to a post.
CO-02	A signed-in user shall be able to retrieve comments for a post in oldest-first order.
CO-03	The system shall keep each post's comment count consistent with saved comments.

Content rules

A post must contain text or an image. Text is limited to 2,000 characters; comments to 1,000. Post images are limited to one JPEG, PNG, or WebP file of up to 3 MB.

The feed should present author information, content, optional image, like count, comment count, and creation time. No feed search, tagging, or bookmarking behavior is specified.

Main Features: Events and Notifications

Event lifecycle

ID	Requirement
EV-01	A signed-in user shall be able to submit an event with title, description, date, time, location, category, and maximum participants.
EV-02	A newly submitted event shall start with pending status.
EV-03	An administrator shall see pending events and approve or reject them.
EV-04	Signed-in users shall see approved events.
EV-05	A student shall be able to join an approved event when space remains, or leave an event they joined.
EV-06	A student shall be able to retrieve the participants of an approved event.

Notifications

ID	Requirement
NO-01	The system shall create notifications for post likes, post comments, event joins, and event review outcomes.
NO-02	A signed-in user shall be able to list their own notifications and unread count.
NO-03	A recipient shall be able to mark one or all of their notifications as read.
NO-04	The server shall emit a new-notification event to the recipient's authenticated connection.

Event states	Only pending events can be reviewed. Approved events are the discoverable event list. Rejected events are not shown there.
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User Stories

Each story corresponds to a capability in the defined scope.

Module	User story
Account	As a visitor, I want to register with my ENAA email, so that I can access the campus space.
Account	As a registered user, I want to sign in and sign out, so that I can control access from my browser.
Account	As a registered user, I want to reset a forgotten password, so that I can regain account access.
Profile	As a signed-in user, I want to edit my details and picture, so that my profile stays current.
Profile	As a signed-in user, I want to view a student's public details, so that I can recognize community members.
Posts	As a signed-in user, I want to publish and read posts, so that I can share campus updates.
Posts	As a signed-in user, I want to like and comment on posts, so that I can respond to others.
Posts	As a student, I want to edit my own post, so that I can correct its text.
Posts	As a post owner, I want to delete my post, so that I can remove content I shared.
Events	As a signed-in user, I want to submit an event, so that it can be considered for publication.
Events	As a student, I want to join or leave an approved event, so that I can manage attendance.
Events	As an administrator, I want to review pending events, so that only reviewed events are published.
Activity	As a recipient, I want to read and clear notifications, so that I can track relevant activity.

Main Pages and Modules

Core website screens are listed separately from supporting service functions.

Screen / module	Purpose
Login	Accept ENAA email and password; provide access to protected pages.
Registration	Create a student account using the required identity and password fields.
Campus feed	Display posts; provide posting and reaction actions.
My profile	Show profile details and authored posts; allow editing and picture upload.
Events	Show approved events; allow submission and student participation.
Admin event review	Show pending submissions and approval or rejection controls.

Supporting modules

Module	Purpose
Authentication service	Issue and verify tokens; load current user; support reset by email.
Post and comment service	Store posts, images, likes, and comments.
Event service	Manage event status, capacity, participation, and participant lists.
Notification service	Persist user-specific activity notices and emit real-time notice events.

Screen boundary This document does not add separate pages for clubs, messages, trips, search, bookmarks, or notifications. Notification handling is specified as a service capability.

Technical Overview

Target technologies and responsibilities for the proposed system.

Layer	Technology / responsibility
Web client	React with JavaScript, React Router, Vite, Tailwind CSS, Axios, and Lucide icons.
API server	Node.js ES modules and Express, with routes, middleware, controllers, and Zod validation.
Database	MongoDB with Mongoose models for users, posts, comments, events, and notifications.
Security	JWT Bearer authentication, bcryptjs password hashing, role checks, and ownership checks.
Media and email	Multer for incoming images, Cloudinary for image storage, and Nodemailer for reset emails.
Live events	Socket.IO for authenticated connections and recipient notification events.
Deployment support	Docker Compose configuration for the backend and MongoDB.

Data at a glance

The principal entities are User, Post, Comment, Event, and Notification. Posts and comments have authors. Events have organizers, participants, and review status. Notifications have recipients, actors, read state, and a related entity.

Integration principle

The client shall submit validated requests to the API and render returned data or errors. The server remains responsible for authentication, authorization, validation, persistence, and business rules.

Session behavior	The browser may store its token for the current session or on the device when the user chooses to remain signed in. Sign-out removes the browser token.
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Main Rules and Requirements

Access and ownership

Rule	Requirement
Authentication	Protected user, post, event, and notification operations require a valid Bearer token.
Roles	Only administrators review pending events. Only students join/leave events, view approved-event participants, or update an owned post.
Ownership	Only the author may update or delete a post. A user may edit only their own profile and mark only their own notifications as read.

Validation and business rules

Area	Requirement
Accounts	Registration requires a unique @enaa.ma email, a 3-100 character name, and an 8-128 character matching password.
Profiles	A profile update requires at least one allowed field. Biography is at most 500 characters; interests at most 20. Profile images are JPEG, PNG, or WebP up to 5 MB.
Events	Date/time must be in the future. Capacity is 1-1,000. Categories: education, sports, workshops, culture, entertainment, trips.
Participation	Joining requires approval and spare capacity. Duplicate joins and leaving without membership are rejected.
Notifications	The system does not create a notification when actor and recipient are the same account.

Basic quality requirements

Layouts shall work on desktop and mobile. The server shall enforce permissions independently of browser controls. Passwords and reset tokens shall be stored as hashes; reset links shall expire after one hour. Sign-out shall clear the browser token, while issued JWTs remain valid until expiry. Errors shall give users clear feedback.

Use Cases

The table captures the main actor, trigger, result, and exception for each journey.

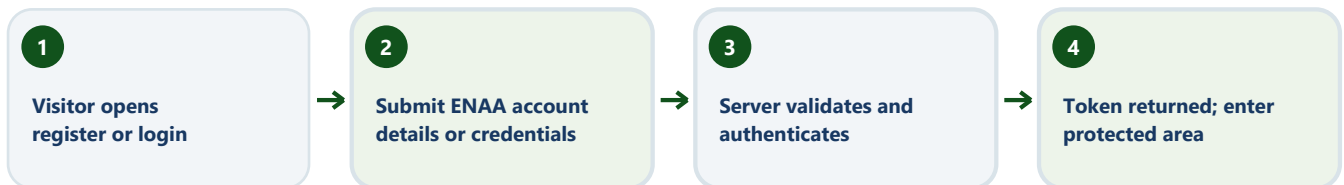
Use case / actor	Main path	Exception or boundary
UC-01 Register / Visitor	Submit valid ENAA details; create student account; receive sign-in token.	Duplicate email or invalid fields are rejected.
UC-02 Sign in / User	Submit credentials; receive token; enter protected area.	Incorrect credentials are rejected.
UC-03 Edit profile / User	Submit allowed fields or picture; save updated profile.	Invalid fields or image are rejected.
UC-04 Publish post / User	Submit text and/or one image; show the new post.	Empty post or invalid image is rejected.
UC-05 React or comment / User	Like/unlike or add a comment; update counts.	Missing post or invalid comment is rejected.
UC-06 Submit event / User	Send event details; save event as pending.	Past time or invalid event fields are rejected.
UC-07 Review event / Admin	Select pending event; approve or reject; notify organizer.	Non-pending event cannot be reviewed.
UC-08 Join / leave event / Student	Join approved event or leave an existing participation.	Full, duplicate, absent, or unavailable event is rejected.
UC-09 Read notices / Recipient	List notices; mark one or all as read.	Another user's notification is inaccessible.

Alternative paths terminate with an error response and do not imply a successful state change. The client should display the returned error when the capability is exposed through a screen.

User Flows

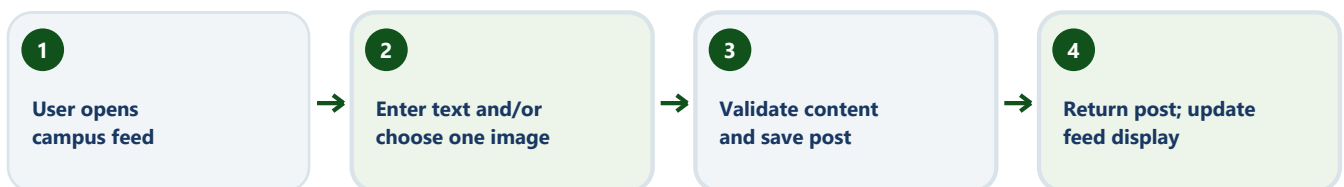
Compact flows show the required order of work and the main decision points.

Account entry



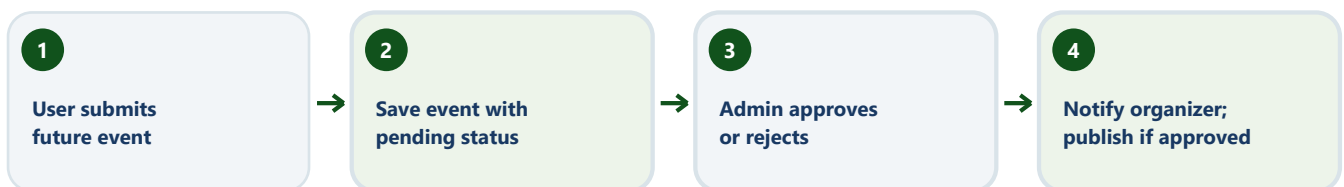
If registration or login fails, show the error and keep the user on the form.

Post creation



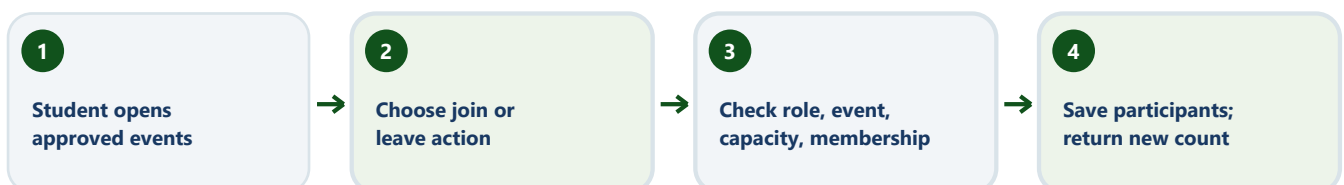
If content is empty or the image is invalid, reject the request and retain the draft.

Event submission and review



Only a pending event can be reviewed; only approved events enter the discoverable list.

Student participation



A join is rejected for a full event or an existing participant. A leave is rejected if the student has not joined.